SIMATS ENGINEERING

CO2 – DRY RUN EVALUATION (ANSWERED)

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1. Breadth-First Search (BFS)

Graph structure: A has children B and C; B has children D and E; C has children F and G. BFS visits each level from left to right.

Iteration	Queue before visit	Visited node	Queue after visit
1	[A]	A	[B, C]
2	[B, C]	B	[C, D, E]
3	[C, D, E]	C	[D, E, F, G]
4	[D, E, F, G]	D	[E, F, G]
5	[E, F, G]	E	[F, G]
6	[F, G]	F	[G]
7	[G]	G	[]

BFS traversal: A → B → C → D → E → F → G

2. Depth-First Search (DFS)

Assuming left-to-right child order, DFS goes deep along the left branch before backtracking.

DFS traversal: A → B → D → E → C → F → G

3. 8-Puzzle – BFS Dry Run

Initial state: 1 3 6 / 5 0 2 / 4 7 8

Goal state: 1 2 3 / 4 5 6 / 7 8 0

BFS expands states in FIFO queue order. The table below shows the first five state expansions using move order Up, Down, Left, Right and avoiding previously visited states. State labels are abbreviated as 3×3 rows separated by "/".

Iter.	Current state	Queue before expansion	Queue after expansion
1	136/502/478	1 states; front 136/502/478	4 states; front 106/532/478
2	106/532/478	4 states; front 106/532/478	5 states; front 136/572/408
3	136/572/408	5 states; front 136/572/408	6 states; front 136/052/478
4	136/052/478	6 states; front 136/052/478	7 states; front 136/520/478
5	136/520/478	7 states; front 136/520/478	8 states; front 016/532/478

1. Which state is expanded in each iteration? The first five are: 1. 136/502/478, 2. 106/532/478, 3. 136/572/408, 4. 136/052/478, 5. 136/520/478.

2. How many states are generated in total? In this BFS run, 500 distinct states were discovered before the goal was reached (including the initial state).

3. Shortest solution path:

Step 0 (Initial): 1 3 6 / 5 0 2 / 4 7 8

Step 1 (Right): 1 3 6 / 5 2 0 / 4 7 8

Step 2 (Up): 1 3 0 / 5 2 6 / 4 7 8

Step 3 (Left): 1 0 3 / 5 2 6 / 4 7 8

Step 4 (Down): 1 2 3 / 5 0 6 / 4 7 8

Step 5 (Left): 1 2 3 / 0 5 6 / 4 7 8

Step 6 (Down): 1 2 3 / 4 5 6 / 0 7 8

Step 7 (Right): 1 2 3 / 4 5 6 / 7 0 8

Step 8 (Right): 1 2 3 / 4 5 6 / 7 8 0

4. Why does BFS find the shortest solution? In an unweighted 8-puzzle, each legal move has equal cost. BFS explores states by increasing depth (number of moves), so the first goal found has the fewest moves.

— End of Answer —